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Learning Multi-Task Robot Trajectory Segmentation from Visual and Kinematic Streams

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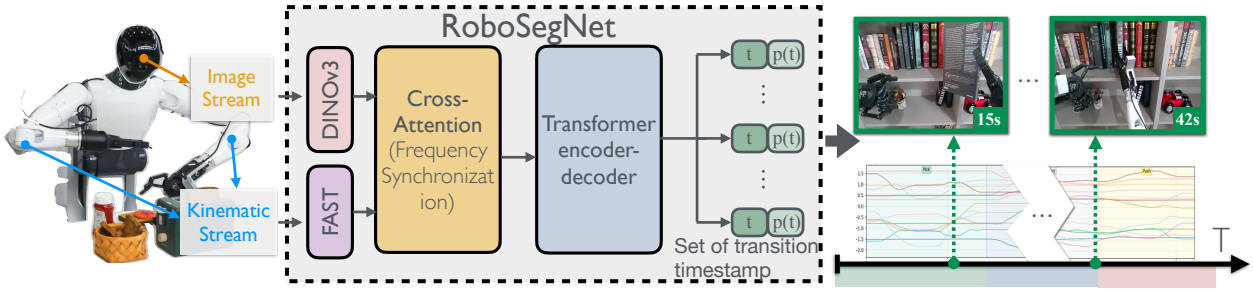


Figure 1. **Overview of RoboSegNet for multimodal trajectory segmentation.** RoboSegNet takes visual and kinematic streams from robot demonstrations (e.g., Insert a book into the bookshelf) to predict a set of transition boundaries. Visual features are extracted using DINOv3 [29], and kinematic features using FAST [24]. RoboSegNet fuses both modalities, and uses a transformer encoder–decoder that outputs transition probabilities $p(t)$ over time t. The colored timeline illustrates aligned visual and joint trajectories, segmented into distinct phases such as Grasp, Insert, and Nudge. Predicted transition points can facilitate robot policy training, such as progress modeling [10, 40].

Abstract

Dividing robot demonstration trajectories into semantically coherent segments can enable efficient policy learning and facilitates adaptation to novel environments. However, existing trajectory segmentation methods using kinematic data are typically designed for single task distribution, and robot trajectories often lack abrupt semantic changes that facilitate visual segmentation methods. We present RoboSegNet, a multi-task robot segmentation framework that jointly learns from visual and kinematic proprioceptive signals. RoboSegNet encodes kinematic inputs using a Discrete Cosine Transform (DCT)-based tokenizer and visual inputs using a visual transformer, fusing them via bidirectional cross-modal attention. RoboSegNet predicts transition boundaries through Hungarian matching. To evaluate RoboSegNet, we introduce RoboSegData, a benchmark derived from the newly released Agibot dataset with densely annotated frame-level transitions. Evaluation suggests that RoboSegNet overcomes the precision challenge of kinematics-based methods, and generalization challenge of visual-based methods, achieving state-of-the-art performance and robust zero-shot performance to unseen scenes, tasks and skills. https://berkeleyautomation.github.io/RoboSegNet/

1. Introduction

Robot demonstration trajectories with synchronized visual and kinematic streams are widely used for robot learning [7, 11, 15]. Temporally segmenting trajectories into shorter, less-variable segments is a classical robot learning problem for analyzing and modeling robot data since it facilitates progress modeling [10, 40], skill reuse [22], and failure recovery [3]. Existing approaches focus on proprioceptive kinematics, assuming that robot trajectories from collected demonstrations lie within a homogeneous task-specific distribution. However, this assumption limits their applicability to datasets for emerging visuomotor policies and Vision–Language–Action (VLA) models, which involve highly diverse scenes, dexterous motions, and multiple distinct tasks. The growing diversity of tasks motivate learning methods that can segment, interpret, and generalize across varied skills and task distributions. In this paper, we address this challenge by jointly leveraging visual and kinematic streams, enabling visual–kinematic fusion to provide richer context for consistent trajectory segmentation across multiple tasks.

We consider a supervised multi-task robot trajectory segmentation problem: given labelled demonstrations collected across multiple manipulation tasks, each comprising synchronized RGB frames and joint-pose trajectories, to learn a

model that can generalize to identify frame-level transition intervals in novel tasks.

Providing fine-grained robot trajectory segmentation is challenging due to ambiguities in both visual semantics and motion patterns: robot trajectories evolve smoothly without abrupt visual semantic changes (e.g., robot gripper occludes the object during interaction); semantically similar task segments may correspond to distinct motions (e.g., "picking up a towel" vs. "picking up the cup"). Therefore, consistent trajectory segmentation across different tasks and scenes requires effectively leveraging information from both visual features and kinematic signals measured by the robot's hardware, such as joint encoders and motors.

To study multi-task robot trajectory segmentation in the context of large-scale demonstration data, we construct RoboSegData, a dataset and benchmark derived from recently released Agibot dataset for large-scale VLA training. Agibot offers large-scale teleoperated demonstrations with observations, robot kinematics, gripper signals, and human-annotated skill primitives spanning 217 tasks in 106 real environments. We convert annotation of Agibot into densely labeled trajectory segments and fine-grained transitions. The RoboSegData comprises 1,210 distinct trajectories, 88,252 transitions, 202 hours total length, covering a wide range of manipulation scenarios with diverse objects, tools, and interaction dynamics.

We propose RoboSegNet, a learned multi-task robot trajectory segmentation framework. RoboSegNet encodes input high-frequency kinematic features with FAST [24], a Discrete Cosine Transform (DCT) [1]-based tokenizer that are typically used for Vision-Language Action models. RoboSegNet fuses both encoded kinematic features and visual features through bidirectional cross-modal attention to form temporally aligned representations. The framework uses a query-based decoder to attend to these features and predict boundary positions and confidences. Training is inspired by DETR [9] which uses Hungarian matching loss to align predictions with ground-truth transitions.

We evaluate RoboSegNet with RoboSegData on on single task, multi-task and out-of-distribution tasks. Evaluation suggests that kinematics-only approaches generalize better to new tasks but are unable to handle complex and visual diverse tasks, while visual-only approaches can localize precise boundaries but sometimes overfit to task-specific visual patterns. RoboSegNet effectively fuse both modalities, achieving state-of-the-art performance on robot trajectory segmentation problem and demonstrating robust zero-shot performance to unseen scenes, tasks and skills.

In this paper, we make three contributions: (1) Identification and formulation of multi-task robot trajectory segmentation problem, (2) RoboSegData, a dataset and benchmark for multi-task robot skill segmentation, (3) RoboSegNet, a multi-task trajectory segmentation framework that jointly learns the representation of visual and kinematic features.

2. Related Work

Robot Trajectory Segmentation Trajectory segmentation is a longstanding problem in robotics, aiming to identify temporal boundaries between motion primitives and their associated piecewise-smooth dynamics [16, 31]. Earlier studies formulations originate from biomechanics and motor control, where human movements were shown to exhibit structured segments such as the "two-thirds power law" in handwriting [36], piecewise-linear arm trajectories [23], and control-theoretic models of primitive decomposition [30]. Recent work explored probabilistic, hierarchical, and reinforcement-learning perspectives for discovering motion primitives [2, 6, 8, 18, 34, 35], including hierarchical Dirichlet Processes [16] and GMM-HMM approaches [17]. These methods mostly rely on kinematic state, assuming smooth, low-dimensional dynamics [14, 33]. While effective for single-task or low variability settings, kinematics-only approaches fail to generalize to modern robot learning datasets where visual semantics, contact events, and object-centric interactions are essential for defining meaningful action boundaries.

Human Action Segmentation in Computer Vision A closely related problem is temporal action segmentation in computer vision, which aims to divide long videos into semantically coherent action units [32]. Transformer-based architectures such as MS-TCN [12], ASFormer [39], and recent models [19, 21, 25, 26, 38, 41] achieve strong performance by modeling fine-grained temporal dependencies. However, directly applying human-action models to robotics is challenging. Robot manipulation trajectories exhibit smooth, non-salient transitions, occlusion during contact, limited pose variation, and dynamics not visible from visual input alone [13, 28]. As a result, visual-only action segmentation struggles with ambiguous motion boundaries in robotic settings, and sometimes unable to generalize across multiple tasks. This gap motivates approaches of RoboSegNet that jointly leverage proprioception and vision, enabling ambiguities in one modality to be resolved by the other.

Trajectory Segmentation for Emerging Robot Learning Algorithms Modern robot learning has emerging capabilities generalized towards complicated and heterogeneous robot tasks such as fixing a bicycle [3], folding cloth [5], making sandwich [4]. These methods benefit from accurate trajectory segment annotation, which enhance learning capability and efficiency through progress modeling[10, 40], skill reuse[22], and failure recovery [3]. At the same time, large robot demonstration datasets, such as BridgeData V2 [37], Open X-Embodiment [11], DROID [15], and Agibot [7], fea-

ture massive visual diversity, heterogeneous tasks, varying embodiments, and inconsistent teleoperation styles. Classical kinematic segmentation methods cannot cope with such variation, while vision-only models often overfit to appearance cues. To address these challenges, we introduce RoboSegData, a benchmark derived from Agibot with dense frame-level skill boundaries, designed specifically to evaluate multi-task trajectory segmentation in complex, real-world robot manipulation scenarios.

3. Problem Definition

Given a robot task demonstration composed of synchronized visual and proprioceptive sequences, the goal of *robot trajectory segmentation* is to partition the demonstration into a sequence of temporally contiguous segments. The segments are typically semantically coherent actions, such as picking up or placing an item.

Definition 3.1 (Robot Trajectory) *We represent each robot demonstration τ as a time-indexed observation sequence where time index $t \in \{1, 2, \ldots, T\}$ denotes a discrete timestep obtained by sampling the continuous trajectory at the visual stream frequency, thereby mapping discrete indices t to real time in seconds. The observation sequence is multi-modal and consisting of visual and kinematic streams:*

$$\tau = \{(I_t, q_t)\}_{t=1}^{T}, \tag{1}$$

where $I_t \in \mathcal{I}$ denotes the visual observation (e.g., an RGB frame) and $q_t \in \mathcal{Q}$ denotes the robot's joint configuration or end-effector pose at time step t. The trajectory length T may vary across demonstrations.

We consider robot trajectory segmentation as a set prediction problem to avoid dense framewise classification of transition versus non-transition states, which will trigger imbalance data issue:

Problem 1 (Robot Trajectory Segmentation) *Given a multimodal robot trajectory τ, the objective is to predict a set of robot transition timestamps $\mathcal{Y} = \{y_1, y_2, \ldots, y_K\}$ that partition τ into K atomic sub-segments:*

$$\tau_k = \tau[y_{k-1} : y_k], \quad \forall y_k \in \mathcal{Y}, \tag{2}$$

where each segment τ_k corresponds to a semantically consistent action, such that all observations in τ_k correspond to the same latent action primitive (e.g., reach*,* grasp*,* lift*), and thus share a coherent high-level intent.*

4. Method

In order to solve the Problem 1, we propose a robot trajectory segmentation framework with multimodal encoders and cross-attention fusion for spatiotemporal transition prediction, as shown in Fig. 2. The model jointly processes RGB and proprioceptive streams and directly predicts a set of transition states using set-based matching.

4.1. Multimodal Feature Fusion

Complex robot demonstrations require understanding both what the robot sees and how it moves. Visual cues capture object interactions and spatial layouts, while proprioceptive signals encode motion dynamics. To enable precise temporal segmentation, we fuse these complementary modalities into a unified representation.

Joint State Features. RoboSegNet encodes the robot's proprioceptive stream using Frequency-space Action Sequence Tokenization (FAST) [24], a discrete cosine transform (DCT)-based action tokenizer originally developed for VLAs. Given a window of continuous joint actions, FAST first normalizes each action dimension using dataset quantiles so that values lie in a fixed range, then applies a per-dimension DCT to map the signal into the frequency domain. DCT coefficients are scaled and rounded so that most high-frequency components become zero, yielding a sparse frequency matrix. This matrix is flattened in a column-first order, placing low-frequency coefficients across all joints first, and then compressed into a short sequence of discrete tokens via byte–pair encoding (BPE) [27].

Visual Features. For the visual stream, we adopt the DINOv3 [29] self-supervised transformer encoder pretrained on large-scale visual data. Given synchronized RGB frames $\{I_t\}_{t=1}^{T}$, DINO extracts per-frame spatial embeddings $F_I \in \mathbb{R}^{T \times d_I}$ that capture object-level and scene-level cues relevant to manipulation and interaction.

Cross Attention. To achieve tight visual–kinematic alignment, we employ a cross-attention fusion module that projects both modalities into a shared latent space $\mathbb{R}^d$. The fused representation is obtained by

$$F_{\text{fuse}} = \text{CrossAttn}(F_q, F_I), \tag{3}$$

where proprioceptive tokens query visual tokens, allowing the model to ground motion states in visual context (e.g., hand–object contact). The fused tokens are temporally ordered to form the input sequence for the transformer backbone.

4.2. Architecture

Backbone. The backbone receives the fused multimodal tokens F_{fuse} and refines them through positional encoding and multi-head self-attention. This produces a global spatiotemporal representation that encodes both task-level and fine-grained motion transitions.

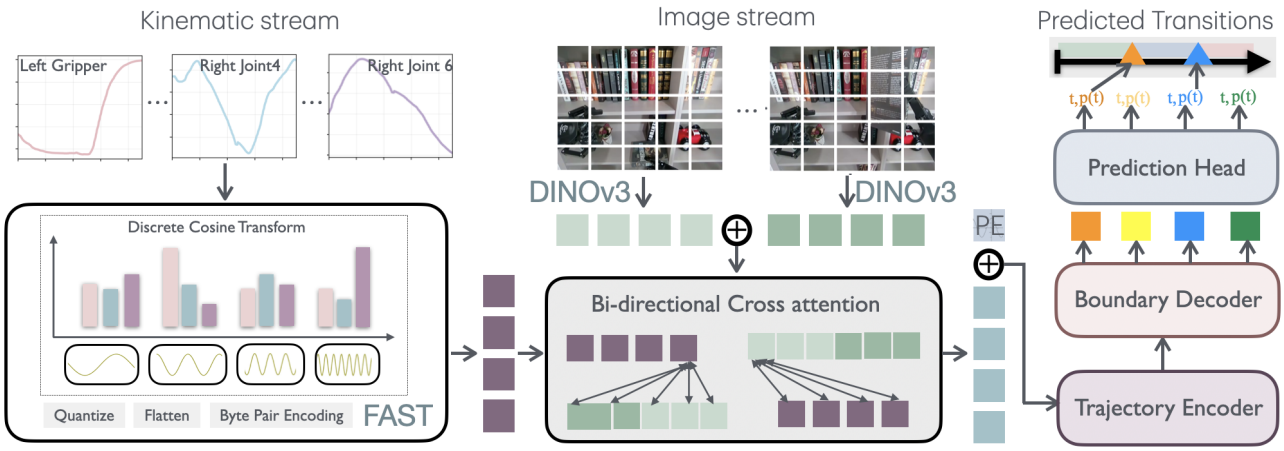


Figure 2. **RoboSegNet** is a multi-task robot trajectory segmentation framework that leverages multimodal input. RoboSegNet encodes proprioceptive input through FAST [14], a discrete cosine transform (DCT) tokenizer that obtains compact frequency-domain motion tokens. It fuses FAST-encoded proprioceptive input and DINOv3 [29]-encoded visual input with cross-modal attention and processes with a transformer encoder–decoder with position encoding (PE) to predict trajectory transition boundaries inspired by DETR [9].

Transformer Encoder. The encoder consists of L_e layers of multi-head self-attention and feedforward blocks that contextualize each timestep with respect to the full trajectory. This design enables bidirectional reasoning, allowing the model to exploit future context when identifying transitions.

Transformer Decoder. The decoder uses a fixed number of learned query embeddings $\{q_i\}_{i=1}^{N}$ that each correspond to a potential transition boundary. Through cross-attention with the encoder outputs, each query predicts the temporal position and confidence of a transition. The output is a sparse set of predicted transition states $\{(\hat{t}_i, \hat{p}_i)\}_{i=1}^{N}$, which are matched to ground truth using Hungarian assignment.

4.3. Loss

Inspired DETR [9], we formulate transition detection as a set prediction problem rather than dense framewise classification of transition versus non-transition states. The key intuition is that transition boundaries in robot trajectories are sparse, unordered, and temporally ambiguous—only a few frames correspond to meaningful skill changes, while most frames reflect smooth continuous motion. Dense labeling thus leads to severe imbalance and uncertain supervision. To address this, each decoder query predicts a potential transition boundary, parameterized by a normalized timestamp and a confidence score. A bipartite matching cost between predictions and ground-truth transitions is computed, and an optimal one-to-one assignment is established via the Hungarian algorithm, ensuring permutation invariance and stable learning under sparse labels. The overall loss combines matching, classification, and regression terms:

$$\mathcal{L} = \mathcal{L}\text{match} + \lambda_1 \mathcal{L}\text{cls} + \lambda_2 \mathcal{L}\text{reg}. \quad (4)$$

Here, $\mathcal{L}$cls encourages confident detection of true transitions, $\mathcal{L}$reg enforces precise temporal localization, and $\mathcal{L}$match integrates both signals for optimal query assignment. This design aligns with DETR's fixed-set prediction philosophy, enabling end-to-end training of the boundary detector without post-processing such as non-maximum suppression. Both λ_1 and λ_2 are hyperparameters.

5. Dataset and Benchmark

5.1. RoboSegData Dataset

We curate RoboSegData from the Agibot dataset [7], a large-scale collection of multimodal robot manipulation trajectories. The latest release contains 1,001,552 teleoperated trajectories totaling 2,976.4 hours, spanning 217 tasks, 87 atomic skills, and 106 scenes across five real-world domains (home, retail, industrial, restaurant, and office). Each episode is collected on a standardized AgiBot G1 humanoid robot with dual 7 Degree-of-Freedom (DoF) arms, interchangeable grippers, and includes synchronized image streams from up to eight cameras (front RGB-D and fisheyes, wrist-mounted RGB-D/fisheye, and rear fisheyes) together with proprioceptive states such as joint and end-effector positions. Trajectories are generated via expert human teleoperation leading to realistic motion variability and contact-rich behaviors in long-horizon tasks (e.g., "make a coffee") rather than isolated atomic motions. The dataset is extensively annotated with sub-task segmentation for each sub-step.

To construct a balanced dataset that covers diverse robot application scenarios and manipulation skills, we sampled a representative subset from the Agibot dataset. Specifically, we selected all tasks containing fewer than 1000 trajectories to ensure feasible training across multiple baselines. As sum-

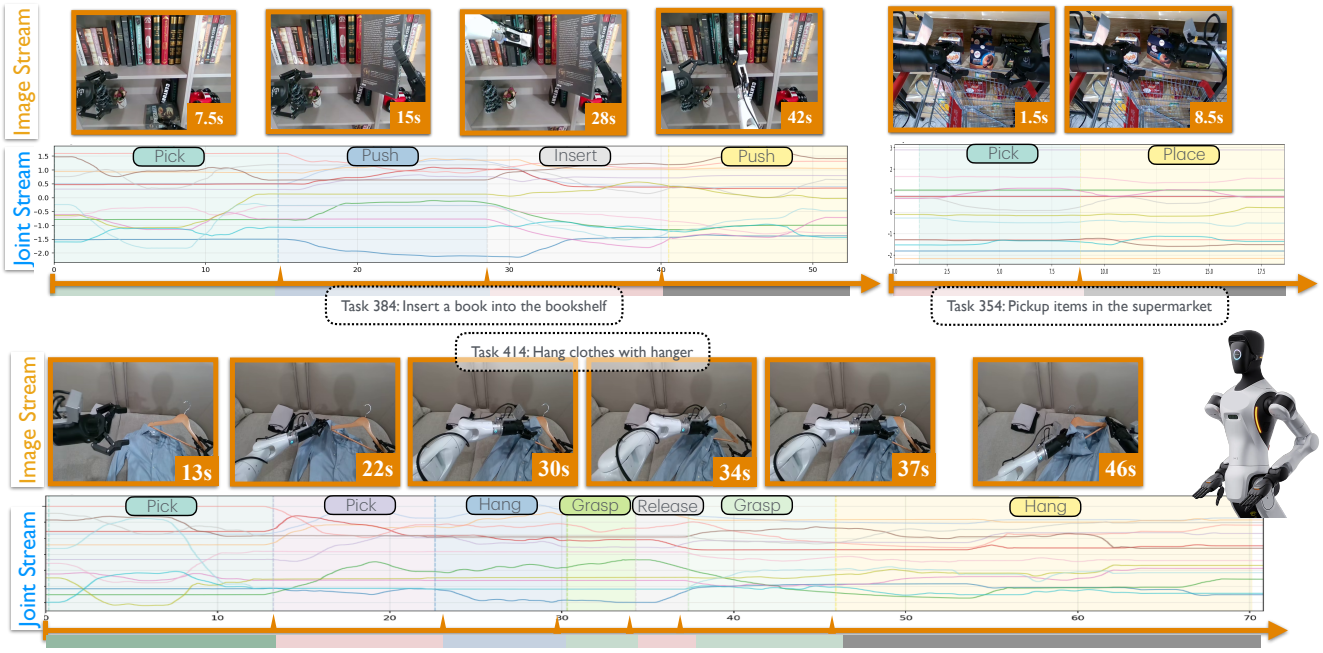


Figure 3. **Example annotated demonstrations in the RoboSegData dataset.** Each trajectory contains synchronized ego-centric image stream and joint stream, annotated with frame-level transition boundaries and fine-grained sub-action labels. In the figure, the top left is Task 384 *Insert a book into the bookshelf*, illustrating transitions such as Pick, Push, and Insert as the robot manipulates objects in a cluttered shelf. Top-right: Task 354 *Pickup items in the supermarket*, showing Pick and Place phases in a visually complex retail environment. Bottom row: Task 414 *Hang clothes with hanger* with transitions such as Pick, Hang, Grasp, and Release.

marized in Table 1, the resulting subset includes 21 tasks, 11,210 trajectories, and 88,252 annotated transitions, totaling 202 hours of demonstrations. This subset spans a wide range of task types, temporal horizons, transition densities, and motion primitives, providing a comprehensive benchmark for evaluating segmentation performance across varied robotic skills. We treat the sub-task annotations as fine-grained segments and convert them into trajectory segments with associated transition timestamps, yielding a dense set of ground-truth boundaries for robot trajectory segmentation, as illustrated in Figure 3.

5.2. Evaluation Metrics

To evaluate the accuracy of predicted transition boundaries, we adopt standard temporal detection metrics at multiple tolerance levels, following prior work in action segmentation and boundary detection.

Boundary Detection Metrics. We first define the time difference error $E_t = |\hat{t} - t^*|$ between each predicted transition $\hat{t}$ and its nearest ground-truth timestamp t^*. A prediction is counted as a true positive if E_t lies within a specified tolerance window n seconds (denoted as $E_t@n$). Using this definition, we compute precision, recall, and F1 scores under different temporal tolerances:

$$\text{F1@}n = \frac{2 \times \text{Precision@}n \times \text{Recall@}n}{\text{Precision@}n + \text{Recall@}n}.$$

Matching between predictions and ground truth is performed using nearest-neighbor rule within each tolerance window. We report F1@1s, F1@2s, and F1@3s as our primary metrics, corresponding to ± 0.5, ± 1, and ± 1.5 seconds tolerance respectively.

Action Segmentation Metrics. Similarly, for evaluating temporal segment quality, we follow standard protocols from human action segmentation. Let $\mathcal{S} = \{\hat{s}_i\}$ be the set of predicted temporal segments and $\mathcal{S}^* = \{s_j^*\}$ be the corresponding ground-truth segments. A predicted segment $\hat{s}_i$ is considered a correct match for s_j^* if their temporal Intersection-over-Union (IoU) satisfies

$$\text{IoU}(\hat{s}_i, s_j^*) \geq \theta,$$

for a given threshold θ. Using this criterion, we compute the *Segmental F1* score:

$$\text{F1@IoU}(\theta) = \frac{2 \cdot \text{Precision}(\theta) \cdot \text{Recall}(\theta)}{\text{Precision}(\theta) + \text{Recall}(\theta)}.$$

To ensure one-to-one correspondence between predicted and ground-truth segments, matching is performed using the

Table 1. **RoboSegData Statistics** The overall dataset of RoboSegData with training, validation and testing involve 11,210 distinct trajectories, 88,252 transitions, 202 hours total length. For the multi-task evaluation, the models are trained on the tasks above the bar, and the out-of-distribution tasks are below the bar. The Task ID maps to the original Task ID in Agibot dataset. The table shows average number of transition (Trans.) and average length (Length) in seconds.

Task ID	Task Name	Trans.	Length	# Traj.	Scenario
374	Sort laundry and personal products	2.0	21.1	91	Industry
398	Sort clothes	5.0	45.0	173	Industry
327	Pickup items in the supermarket	6.0	41.0	209	Supermarket
385	Pickup in the produce section	6.0	41.7	214	Supermarket
389	Pickup in the supermarket	4.0	39.2	312	Supermarket
376	Sort electronic products	5.0	41.5	403	Industry
375	Brew tea	8.0	68.2	422	Home
357	Wash dishes with dishwasher	16.0	123.5	428	Home
358	Toast bread	5.0	40.4	470	Home
356	Packing in the supermarket	5.0	32.1	475	Supermarket
354	Pickup items in the supermarket	2.0	20.2	516	Supermarket
360	Packing in the supermarket	7.0	39.7	517	Supermarket
372	Packing in the supermarket	9.0	58.6	537	Supermarket
359	Sort in the warehouse	5.0	42.9	644	Industry
378	Clear table in the restaurant	12.0	66.3	739	Restaurant
390	Checkout and scan barcode	11.0	72.5	750	Supermarket
377	Packing in e-commerce	6.0	85.1	794	Industry
384	Insert a book into the bookshelf	4.0	46.8	808	Home
380	Packing in e-commerce	8.0	96.5	815	Industry
424	Clear the countertop waste	9.0	61.6	916	Home
392	Brush water bottle	14.2	123.8	977	Home

Hungarian algorithm under each threshold θ. We report Seg-F1@10%, Seg-F1@25%, and Seg-F1@50% as our primary segment-level metrics.

6. Experiment

In this section, we study the effectiveness of proposed RoboSegNet framework by evaluating on RoboSegData.

6.1. Evaluation Setup

Model Setup RoboSegNet uses off-the-shelf FAST+ action tokenizer, which is trained on 1 million real robot action sequences with a variety of robot embodiments [24], but it is not trained on Agibot. The tokenizer is connected a single-layer transformer with 4 attention heads operating on 600-dimensional token embeddings, with embedding dropout of 0.2 and transformer dropout of 0.2. The visual encoder employs a 2-layer temporal transformer processing the 9 sampled frames, projecting each 1024-dimensional DINOv3 feature to a 256-dimensional hidden representation with dropout rate 0.1. Cross-modal fusion is achieved through bi-directional attention with 8 attention heads in a 512-dimensional space. The encoder and decoder each consist of 6 transformer layers with hidden dimension 512, 8 attention heads, feedforward dimension 2,048. The model learns learned boundary queries at 1Hz that predict both positions and confidence scores for potential task boundaries.

Training Setup All models are trained using the AdamW optimizer [20], which has become the standard choice for transformer-based architectures. We use a warmup schedule with 500 linear warmup steps starting from 10^{-7}, followed by cosine annealing decay to a minimum learning rate of 10^{-6} over the training duration. Training proceeds for up to 200 epochs with early stopping based on validation F1 score, using a patience of 10 epochs.

Baseline Setup We consider different kinds of state-of-the-art robot trajectory segmentation baselines from robotics and computer vision community. For methods using only proprioceptive data, we consider Transition State Clustering [16] and Gaussian Mixture Model-Hidden Markov Model (GMM-HMM) used by SWIRL [17]. We set the confidence threshold by performing a parameter sweep from 0.1 to 0.9 and use the highest confidence threshold (0.1). We setup MS-TCN [12] and ASFromer [39] in the same structure as the original repository and performed hyperparameter sweep across different learning rates and depths. In the evaluation, we use 2 stages with 5 layers with learning rate of 5×10^{-4}, Binary Cross Entropy (BCE) loss and confidence threshold of 0.2. We setup ASFormer with 10 encoding layers, 4 decoder stages and 5 decoder layers.

Ablation Setup We setup a set of ablation studies of RoboSegNet given different modalities and information. We ablate RoboSegNet with no visual input (FAST [24]) and no proprioceptive input (DINOv3 [29]). We use the same depth of multi-layer perceptron architecture with mean squared error loss on dense boundary labels. We use Multi-Layer Perceptron (MLP) to connect the features and use Mean Squared Error (MSE) as the loss. All baseline models are trained with the same hyperparameters and data splits as RoboSegNet.

6.2. Singe Task Performance

We evaluation RoboSegNet all the training tasks in RoboSegData. For each task, we train a separate model and evaluate using only demonstrations from that task.

Compared to Alternative Methods Table 2 compares the performance of RoboSegNet against other state-of-the-art baseline methods and ablations the single-task benchmark. The unsupervised, proprioceptive-only methods (TSC [16] and SWIRL [17]) demonstrate low precision in identifying the transition points; for example, SWIRL scores only 2.47% on F1@2s and 33.06% on Seg-F1@50%. This finding suggests that conventional approaches that do not rely on visual semantics and purely use proprioceptive data is not sufficient for segmenting dexterous and complex tasks in RoboSegData. The supervised, visual-centric baselines (MS-TCN [12] and ASFormer [39]) achieve significantly stronger

Table 2. **Single Task Performance and Ablation Studies**. We compare RoboSegNet with alternative methods and ablations. For proprioceptive-only baseline, we include Gaussian Mixture Model-Hidden Markov Model (GMM-HMM) and Transition State Clustering (TSC); both of the methods are unsupervised. For visual-only baseline, we include Multi-Stage Temporal Convolutional Network (MS-TCN) [12] and ASFormer [39].

Method	F1@1s (%)	F1@2s (%)	F1@3s (%)	Seg-F1@10% (%)	Seg-F1@25% (%)	Seg-F1@50% (%)
TSC* [16]	3.96	5.51	6.63	48.56	43.94	31.57
SWIRL* [17]	1.69	2.47	3.10	56.75	53.39	33.06
MS-TCN [12]	15.88	23.77	29.06	70.75	69.82	53.67
ASFormer [39]	**53.17**	63.63	67.37	86.34	85.87	81.46
FAST [24]	21.41	25.54	27.51	48.82	46.21	21.55
DINOv3 [29]	21.49	25.76	26.64	45.09	38.26	17.72
FAST+DINOv3	23.52	27.80	29.03	49.12	44.94	25.78
RoboSegNet	51.54	**68.95**	**73.82**	**89.51**	**89.25**	**86.45**

Table 3. **In-Distribution Performance of RoboSegNet Trained on In-Distribution Tasks of RoboSegData**. The task distribution covers a variety of scenarios and skills.

Method	F1@1s (%)	F1@2s (%)	F1@3s (%)	Seg-F1@10% (%)	Seg-F1@25% (%)	Seg-F1@50% (%)
MS-TCN [12]	13.29	17.58	20.09	75.05	74.38	67.04
ASFormer [39]	15.04	19.70	22.62	75.00	74.11	66.08
FAST [24]	**21.95**	29.69	34.14	58.33	57.40	38.66
DINOv3 [29]	19.35	22.12	22.14	40.18	29.10	13.04
RoboSegNet	20.66	**38.69**	**50.14**	**84.83**	**83.86**	**74.62**

results. ASFormer performs well with an F1@1s score of 53.17% (vs. 51.54% for RoboSegNet). However, RoboSegNet substantially outperforms ASFormer as the temporal tolerance increases (F1@2s: 68.95% vs. 63.63%; F1@3s: 73.82% vs. 67.37%) and achieves a decisive lead in segmental F1 scores (Seg-F1@50%: 86.45% vs. 81.46%). This suggests that while visual-only methods like ASFormer are highly adept at precise boundary localization in recognizing the visual patterns given a single task, the joint visual-kinematic fusion in RoboSegNet enables it to produce more semantically coherent and holistically accurate segments

Ablation Studies The second half of the Table 2 presents the ablation of RoboSegNet to evaluate the core design choices of RoboSegNet. The full RoboSegNet achieves strong results with 68.95% F1@2s, 89.51% F1@3s, and 86.45% Seg-F1@50%. An important finding for the ablation is the importance of multimodal fusion. Both single-modality variants show performance degradation: Without visual input and using only FAST encoder [24], RoboSegNet achieves only 25.54% F1@2s, and using only DINOv3 [29] achieves just 25.76% F1@2s. Additionally, RoboSegNet's architecture is essential for effective modality fusion. Using an MLP to combine the features with MSE as the loss fails to significantly improve performance over either single-modality variant.

Table 4. **Zero-Shot Performance of RoboSegNet on 3 Out-of-Distribution Tasks**. The out-of-distribution tasks are packing in e-commerce, clear the countertop waste, and brush water bottle.

Method	F1@1s (%)	F1@2s (%)	F1@3s (%)	Seg-F1@10% (%)	Seg-F1@25% (%)	Seg-F1@50% (%)
MS-TCN [12]	5.41	7.31	8.61	72.67	72.08	66.87
ASFormer [39]	5.65	8.28	10.09	75.76	75.48	71.47
DINOv3 [29]	17.39	19.87	19.87	38.39	25.90	11.62
FAST [24]	**18.16**	24.46	28.34	53.39	52.78	32.13
RoboSegNet	16.45	**30.31**	**39.41**	**82.64**	**81.89**	**74.83**

6.3. Multi-Task Performance

To evaluate RoboSegNet's ability to learn generalizable representations across diverse manipulation tasks, we train a single model on all the training data simultaneously and evaluate both in-distribution performance on the training tasks and out-of-distribution generalization to another three held-out tasks.

In-Distribution Performance Table 3 presents the performance of the multi-task RoboSegNet model on the 18 training tasks. This multi-task setting is significantly more challenging for all methods. For example, for MS-TCN [12] and ASFormer [39], which were strong in the single-task scenario, drop from 63.63% F1@2s in the single-task evaluation to just 19.70% in the multi-task setting. This is because, in a single-task setup, robot skills may present different granularities as visual patterns for different tasks. For single task, the scenarios are relatively consistent to recognize, so that visual-only models can successfully recognize the transition points. However, these models tend to overfit in the presence of high visual diversity of 18 different tasks, skills, and object interactions. In contrast, RoboSegNet demonstrates stronger ability to generalize, achieving an F1@2s of 38.69% and an F1@3s of 50.14%. RoboSegNet can achieve a Seg-F1@50% of 74.62% that decisively outperforming all other methods (e.g., 66.08% for ASFormer). This suggests the joint fusion of visual and proprioceptive data is uniquely capable of learning a generalizable representation that produces holistically coherent segments across varied tasks, rather than just memorizing task-specific visual cues.

Out-of-Distribution Performance Table 4 evaluates the multi-task model's generalization to three previously unseen tasks: packing in e-commerce, clearing countertop waste, and brushing water bottles. These tasks introduce novel objects and motion patterns and environmental contexts not seen during training. RoboSegNet demonstrates robust zero-shot transfer, achieving 30.31% F1@2s and 74.83% Seg-F1@50%. This performance, which is comparable in coherence (Seg-F1) to the in-distribution results, which suggests that the model learns generalizable representations of

Table 5. **Confidence Threshold Sensitivity Analysis**. Performance across different confidence thresholds (Confid.) on in-distribution and out-of-distribution tasks. RoboSegNet achieves robust performance in the presence of variable confidence thresholds.

	In-Distribution				Out-of-Distribution			
Confid.	F1@1s (%)	F1@2s (%)	F1@3s (%)	Seg-F1@50% (%)	F1@1s (%)	F1@2s (%)	F1@3s (%)	Seg-F1@50% (%)
0.1	19.05	35.32	42.29	41.75	17.69	32.64	39.37	43.44
0.2	19.40	36.04	43.26	44.09	17.98	33.20	40.20	46.43
0.3	19.65	36.53	43.97	45.78	18.19	33.67	40.87	48.73
0.4	19.93	37.16	44.84	47.92	18.39	34.08	41.49	51.26
0.5	20.42	38.05	46.06	51.30	18.66	34.66	42.32	54.76
0.6	20.94	39.01	47.63	56.60	18.98	**35.19**	43.15	59.50
0.7	**21.25**	**39.70**	49.53	65.08	**19.00**	35.09	**43.98**	67.33
0.8	20.66	38.69	**50.14**	**74.62**	16.45	30.31	39.41	**74.83**
0.9	2.48	4.34	5.57	71.46	0.44	0.90	1.24	73.47

skill boundaries rather than merely memorizing task-specific patterns. In contrast, the capacity for generalization stands in stark contrast to the visual-only baselines. Both MS-TCN and ASFormer, which overfit heavily in the multi-task setting, fail completely at zero shot performance to novel tasks and scenes, with F1@2s scores of only 7.31% and 8.28%, respectively. The ablation also suggests similar pattern: kinematic-only variant (FAST) achieves 24.46% F1@2s, substantially outperforming the visual-only variant (DINOv3) at 19.87% F1@2s. This suggests that proprioceptive motion patterns provide a more transferable signal across tasks than visual features, which appear to be more domain-specific.

Discussion Evaluation data suggests that segmenting large-scale, multi-task robot demonstrations present a unique challenge to existing visual-only and kinematic-only models due to the complexity and diversity of large-scale, multi-task robot demonstrations. For example, visual-only methods such as ASFormer [39] achieve strong performance in single-task settings and at precise boundary localization, but their reliance on appearance cues makes it difficult to maintain coherent segments or generalize across the full diversity of tasks. Similarly, kinematic-only approaches capture transferable motion patterns but lack the semantic grounding needed for fine-grained segmentation in complex scenes. These observations highlight the opportunity for multi-modal fusion methods, such as RoboSegNet, that effectively leverage both streams to achieve robust, scalable segmentation performance across multiple tasks.

6.4. Sensitivity

One of the key parameters in RoboSegNet is the confidence threshold used to filter transition predictions from the query decoder. This threshold controls the trade-off between recall and precision when determining valid transition boundaries. A lower threshold increases recall but introduces more false positives, while a higher threshold enforces stricter confidence but risks missing true transitions. Tab. 5 reports performance across thresholds from 0.1 to 0.9 for both in-distribution and out-of-distribution tasks. We observe that performance steadily improves from 0.1 to 0.7, reaching its peak at 0.7 with F1@2s $= 39.70\%$ and Seg-F1@50% $= 65.08\%$ on in-distribution tasks, and F1@2s $= 35.09\%$ and Seg-F1@50% $= 67.33\%$ on OOD tasks. This indicates that moderate confidence filtering effectively balances temporal localization accuracy and boundary consistency. For threshold larger than 0.8, however, both F1 and Recall drop sharply, suggesting over-pruning of valid boundaries. Even when the boundary count decreases, segment-level F1@50% remains relatively high, suggesting that the remaining transitions are temporally well-aligned and semantically consistent. Sensitivity evaluation suggests that RoboSegNet shows robustness across a wide range of confidence thresholds, which are typically sensitive to DETR-based detection models.

7. Conclusion

In this paper, we introduced the problem of multi-task robot trajectory segmentation from synchronized visual and kinematic streams, addressing the limitations of single-task segmentation methods that fail to generalize across diverse manipulation scenarios. To study this problem, we constructed RoboSegData, a large-scale benchmark derived from the Agibot dataset, featuring densely annotated frame-level transitions across 21 manipulation tasks. We further proposed RoboSegNet, a transformer-based segmentation framework that performs joint visual–kinematic fusion via bidirectional cross-modal attention and predicts transition boundaries in an end-to-end manner through set-based matching. Experimental results suggest that multimodal fusion is essential for learning transferable transition representations across tasks, and the model trained with both visual and proprioceptive modalities exhibits strong zero-shot generalization to unseen manipulation tasks.

Limitations While RoboSegNet demonstrates strong performance across single-task, multi-task, and zero-shot settings, we acknowledge that our study focuses on Agibot where dense, high-quality annotations are available. This enables controlled benchmarking but does not fully reflect the heterogeneity of robot datasets such as Open X-Embodiment [11]. In those datasets, making consistent segmentation annotation is challenging even for human annotators. We consider it as future work to extend RoboSegNet to operate under weak, partial, or self-supervised supervision for those datasets.

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